Target Localization and Measurement Association in PCL-PET Hybrid Heterogeneous Network

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This article discusses the problem of target localization and measurement association in a hybrid heterogeneous network consisting of passive coherent location (PCL) and passive emitter tracking (PET). The distinction from existing literature lies in this article's consideration of the general scenario where PCL receivers can only receive signals from specific transmitters. Two target localization algorithms are proposed based on semidefinite relaxation and Newton–Homotopy methods, corresponding to situations where the initial target position is unknown and known, respectively. Furthermore, this article introduces a measurement association algorithm for the PCL-PET hybrid heterogeneous network. It demonstrates that the test statistic follows a chi-square distribution, providing theoretical guidance for selecting thresholds in hypothesis decisions. Monte Carlo simulations indicate that the proposed localization algorithms achieve accuracy close to the Cramér–Rao lower bound, and the measurement association algorithm effectively discerns whether measurements originate from the same target. In addition, contour plots of localization accuracy illustrate the more significant potential of the PCL-PET hybrid heterogeneous network compared to single PCL and PET networks.

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I. INTRODUCTION

Multisensor networking is an effective means to improve target localization accuracy [1], [2], [3] and expand the detection range [4], [5]. Considering that various sensors with different characteristics can be complementary [6], [7], [8], in recent years, more and more research is no longer limited to homogeneous networks composed of the same type of sensors, and heterogeneous networks have gradually become a research hotspot [9], [10], [11]. Passive coherent localization (PCL) and passive emitter tracking (PET) both belong to the passive detection system. The hybrid heterogeneous network of PCL-PET not only retains radio-silent properties but also achieves higher accuracy, thus attracting significant attention [12], [13], [14]. In this article, we discuss the PCL-PET hybrid heterogeneous network, where PCL measures the bistatic distance, and PET measures the time difference of arrival. By plotting Cramér–Rao lower bound (CRLB) contour lines, we demonstrate that the PCL-PET hybrid heterogeneous network achieves higher target localization accuracy than PCL and PET networks. We focus on the target localization and measurement association problems within this heterogeneous network.

Target localization is a longstanding problem characterized by solving a system of nonlinear equations regarding the target's position. Existing algorithms can be categorized into iterative and noniterative approaches. Iterative algorithms include the Taylor series method [15], [16], which involves expanding the nonlinear equation system around an initial estimate of the target position and solving it iteratively based on measurement information. The least squares-based iterative algorithm [17], [18], [19] transforms the localization problem into a constrained least squares (CLS) problem by introducing auxiliary variables and iteratively solving it. Semidefinite programming (SDP) algorithm [20], [21], [22] utilizes semidefinite relaxation to convert the localization problem into a convex optimization problem, which can then be solved using the interior point method. Noniterative algorithms include spherical intersection algorithm, spherical interpolation algorithm [23], [24], [25], etc. Although the accuracy of these algorithms is difficult to reach the CRLB, they provide an explicit solution approach for the localization problem and are easy to implement. Chan and Ho [26] proposed a two-step estimation algorithm (TSE). In the first step, the constraint relationship between auxiliary variables and the target position is disregarded to compute the initial localization result. Then, the second step improves the localization result using the constraint relationship. The accuracy of TSE can approach the CRLB under conditions of small measurement errors, making it an algorithm that balances computational complexity and localization accuracy. Many extension algorithms were generated based on TSE [27], [28], [29].

It is worth noting that the aforementioned target localization algorithms are developed in homogeneous networks and do not consider heterogeneous network scenarios. In addition, existing algorithms typically assume that measurements between any transmitter and receiver in the PCL network are available. However, in more general scenarios, such as when the target is missed detection or when transmitters and receivers operate on different frequency bands, this can result in only specific transmitter–receiver pairs being able to obtain target measurements or receivers only receiving signals from specific transmitters. While Aubry et al. [17] extended TSE, CLS, and iterative constrained least squares (ICLS) algorithms for the PCL-PET hybrid heterogeneous network, achieving results close to the CRLB under small measurement errors, they still not applicable to the general localization scenarios mentioned above. In this article, we propose two localization algorithms tailored for general localization scenarios, corresponding to situations where the initial estimate of the target position is available and unavailable.

Measurement association is crucial in the radar network for distinguishing whether different measurements originate from the same target, which is a step in eliminating false tracks [30], [31], [32]. Due to differences in the signals emitted by different targets, the association between PET measurements is considered known. However, the correspondence between PCL measurements and between PCL and PET measurements requires additional determination. This article proposes a measurement association algorithm for the PCL-PET hybrid heterogeneous network, which fully considers the scenario where PET and PCL measurement errors are not identical.

The main contributions of this article are as follows.

1) We develop a target localization algorithm for PCL-PET hybrid heterogeneous networks without initial target position information. Initially, the localization problem is formulated as a quadratically constrained quadratic problem (QCQP), which is nonconvex and challenging to solve. Subsequently, by performing semidefinite relaxation, an SDP problem is obtained, which can be directly solved using the CVX toolbox. Finally, a linear equation set regarding localization error is obtained through first-order Taylor expansion. Higher precision localization result is achieved after error correction. Simulation results demonstrate that the algorithm can reach the CRLB in the presence of significant measurement errors.
2) A target localization algorithm is proposed for cases where the initial value of the target position is known. We assume that the initial estimate of the target position can be obtained through certain means but with significant error. It is well known that when the initial guess error is large, the Newton–Raphson method may diverge. This article proposes a target localization algorithm based on the idea of Newton–Homotopy method [33], [34] to address this problem. Newton–Homotopy is an improved algorithm that exhibits more robust global convergence and better numerical stability. Through simulations, we demonstrate its advantages over the Newton–Raphson method in the case of larger initial errors.
3) A measurement association algorithm is proposed for PCL-PET hybrid heterogeneous networks. After obtaining the localization result of a specific measurement combination, the residual of the measurement equation is calculated, and a test statistic is constructed based on this residual to determine the correctness of the measurement combination. This article demonstrates that the test statistic follows the chi-square distribution when the measurement combination is true, providing theoretical guidance for selecting decision thresholds.

The rest of the article is organized as follows. Section II introduces the PCL-PET hybrid heterogeneous network considered in this article. Section III proposes two target localization algorithms, as well as the measurement association algorithm. Section IV conducts simulation verification for the algorithms proposed in this article. Finally, Section V summarizes the work.

Notations: Scalars are denoted by italic letters; bold lower-case and upper-case letters correspond to vectors and matrices. The symbols $\|\cdot\|$, $\odot$, $(\cdot)^{\mathrm{T}}$, $\mathrm{tr}(\cdot)$, $E\{\cdot\}$, and $\mathrm{rank}(\cdot)$stand for the ℓ_2 norm, the elementwise product, the transposition, the trace, the expectation, and the rank operation, respectively. $\mathrm{diag}(\cdot)$ represents constructing the diagonal matrix or extracting the diagonal elements of the matrix. $\mathrm{blkdiag}(\cdot)$ represents constructing the block diagonal matrix. $[\boldsymbol{A}]_{i,j}$ and $[\boldsymbol{a}]_{i:j}$ are the (i, j)th element of $\boldsymbol{A}$ and the subvector with the i th to the j th elements of $\boldsymbol{a}$, respectively. $[\boldsymbol{A}]_{i:j,m:n}$ represents the submatrix of $\boldsymbol{A}$. $\mathbf{O}_{m\times n}$ and $\mathbf{0}_n$ are $m \times n$ zero matrices and all-zero vector with length n. $\boldsymbol{A} \succcurlyeq 0$ means that $\boldsymbol{A}$ is a positive semidefinite matrix.

II. PROBLEM FORMULATION

Assume the existence of a PCL-PET hybrid heterogeneous network in an N_{D}-dimensional space, where the PCL network consists of N_{PCL} transmitters and N_{PCL} receivers, and the PET network consists of N_{PET}receiver nodes and one reference node. Existing literature often assumes that any transmitter–receiver pair in the PCL network can form measurements, while Aubry et al. [17] further assumed that PET nodes share the same coordinates as PCL receivers. The advantage of this assumption is that it can reduce the number of auxiliary variables needed for introduction, facilitating the solution. However, the assumption mentioned above may be difficult to satisfy due to missed detection, inconsistent frequency bands between transmitters and receivers, and station deployment restrictions. Therefore, a more general scenario is considered in this article.

This article considers that PCL receivers can only receive signals from the transmitter with the same numbering. The coordinates of PET receiver nodes can be arbitrarily set and do not need to be consistent with those of PCL receivers. Let the coordinates of the ith transmitter and receiver be denoted as $\boldsymbol{t}_i$ and $\boldsymbol{s}_i$, and the coordinates of the reference node and jth receiver node can be denoted as $\boldsymbol{p}$ and $\boldsymbol{q}_j$, respectively. In the PCL-PET hybrid heterogeneous network, the measurements of the target at position $\boldsymbol{u}^0$ are defined as follows:

$$r_{\mathrm{PCL},i} = \left\|\boldsymbol{u}^0 - \boldsymbol{t}_i\right\| + \left\|\boldsymbol{u}^0 - \boldsymbol{s}_i\right\| \tag{1}$$

$$\tilde{r}_{\mathrm{PCL},i} = r_{\mathrm{PCL},i} + \delta_{\mathrm{PCL},i},\ i = 1, 2, \ldots, N_{\mathrm{PCL}} \tag{2}$$

$$r_{\mathrm{PET},j} = \left\|\boldsymbol{u}^0 - \boldsymbol{q}_j\right\| - \left\|\boldsymbol{u}^0 - \boldsymbol{p}\right\| \tag{3}$$

$$\tilde{r}_{\mathrm{PET},j} = r_{\mathrm{PET},j} + \delta_{\mathrm{PET},j},\ j = 1, 2, \ldots, N_{\mathrm{PET}} \tag{4}$$

where $\delta_{\mathrm{PCL},i}$ and $\delta_{\mathrm{PET},j}$ represent measurement errors, which follow zero mean Gaussian distributions. The definitions of $\boldsymbol{\delta}_{\mathrm{PCL}}$ and $\boldsymbol{\delta}_{\mathrm{PET}}$ are in (7) and (8), and the corresponding covariance matrices are $\boldsymbol{Q}_{\mathrm{PCL}}$ and $\boldsymbol{Q}_{\mathrm{PET}}$. By setting coordinates appropriately, other scenarios can be obtained. For example, when setting $\boldsymbol{s}_1 = \boldsymbol{s}_2 = \ldots = \boldsymbol{s}_{N_{\mathrm{PCL}}}$, a multitransmitter single-receiver PCL network can be obtained. This implies that the localization algorithm derived in this article is still applicable in other scenarios. To simplify subsequent expressions, let

$$\boldsymbol{r}_{\mathrm{PCL}} = \left[r_{\mathrm{PCL},1}, \ldots, r_{\mathrm{PCL},N_{\mathrm{PCL}}}\right]^{\mathrm{T}} \tag{5}$$

$$\boldsymbol{r}_{\mathrm{PET}} = \left[r_{\mathrm{PET},1}, \ldots, r_{\mathrm{PET},N_{\mathrm{PET}}}\right]^{\mathrm{T}} \tag{6}$$

$$\boldsymbol{\delta}_{\mathrm{PCL}} = \left[\delta_{\mathrm{PCL},1}, \ldots, \delta_{\mathrm{PCL},N_{\mathrm{PCL}}}\right]^{\mathrm{T}} \tag{7}$$

$$\boldsymbol{\delta}_{\mathrm{PET}} = \left[\delta_{\mathrm{PET},1}, \ldots, \delta_{\mathrm{PET},N_{\mathrm{PET}}}\right]^{\mathrm{T}}. \tag{8}$$

The target localization problem can be concluded as estimating $\boldsymbol{u}^0$ using $\tilde{r}_{\mathrm{PCL},i}$ and $\tilde{r}_{\mathrm{PET},j}$ under the condition of known $\boldsymbol{t}_i$, $\boldsymbol{s}_i$, $\boldsymbol{p}$, and $\boldsymbol{q}_j$, where $i = 1, 2, \ldots, N_{\mathrm{PCL}}$ and $j = 1, 2, \ldots, N_{\mathrm{PET}}$. The measurement association problem can be concluded as determining whether a set of measurements comes from the same target. In Section III, we develop corresponding algorithms for these two problems.

III. TARGET LOCALIZATION AND MEASUREMENT ASSOCIATION ALGORITHMS

A. SDP-Based Localization Algorithm

Let us rearrange (1) as

$$r_{\mathrm{PCL},i} - \left\|\boldsymbol{u}^0 - \boldsymbol{s}_i\right\| = \left\|\boldsymbol{u}^0 - \boldsymbol{t}_i\right\| \tag{9}$$

and then squaring both sides. After rearrangement, yields

$$\frac{1}{2}\left({r_{\mathrm{PCL},i}}^2 + \|\boldsymbol{s}_i\|^2 - \|\boldsymbol{t}_i\|^2\right) - (\boldsymbol{s}_i - \boldsymbol{t}_i)^{\mathrm{T}}\boldsymbol{u}^0 - r_{\mathrm{PCL},i}L_{\mathrm{PCL},i} = 0. \tag{10}$$

Similarly, for (3), it can be obtained that

$$\frac{1}{2}\left({r_{\mathrm{PET},j}}^2 + \|\boldsymbol{p}\|^2 - \left\|\boldsymbol{q}_j\right\|^2\right) - \left(\boldsymbol{p} - \boldsymbol{q}_j\right)^{\mathrm{T}}\boldsymbol{u}^0 + r_{\mathrm{PET},j}L_{\mathrm{PET}} = 0 \tag{11}$$

where $L_{\mathrm{PCL},i} = \|\boldsymbol{u}^0 - \boldsymbol{s}_i\|$ and $L_{\mathrm{PET}} = \|\boldsymbol{u}^0 - \boldsymbol{p}\|$. Stacking (10) and (11) in the PCL-PET hybrid heterogeneous network yields the following matrix equation:

$$\boldsymbol{h} - \boldsymbol{G}\boldsymbol{\theta}^0 = \mathbf{0}_{N_{\mathrm{PCL}}+N_{\mathrm{PET}}} \tag{12}$$

where

$$\boldsymbol{h} = \frac{1}{2}\begin{bmatrix} {r_{\mathrm{PCL},1}}^2 + \|\boldsymbol{s}_1\|^2 - \|\boldsymbol{t}_1\|^2 \\ \vdots \\ {r_{\mathrm{PCL},N_{\mathrm{PCL}}}}^2 + \left\|\boldsymbol{s}_{N_{\mathrm{PCL}}}\right\|^2 - \left\|\boldsymbol{t}_{N_{\mathrm{PCL}}}\right\|^2 \\ {r_{\mathrm{PET},1}}^2 + \|\boldsymbol{p}\|^2 - \left\|\boldsymbol{q}_1\right\|^2 \\ \vdots \\ {r_{\mathrm{PET},N_{\mathrm{PET}}}}^2 + \|\boldsymbol{p}\|^2 - \left\|\boldsymbol{q}_{N_{\mathrm{PET}}}\right\|^2 \end{bmatrix} \tag{13}$$

$$\boldsymbol{G} = \begin{bmatrix} (\boldsymbol{s}_1 - \boldsymbol{t}_1)^{\mathrm{T}} & r_{\mathrm{PCL},1} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ (\boldsymbol{s}_{N_{PCL}} - \boldsymbol{t}_{N_{PCL}})^{\mathrm{T}} & 0 & \cdots & r_{\mathrm{PCL},N_{\mathrm{PCL}}} & 0 \\ (\boldsymbol{p} - \boldsymbol{q}_1)^{\mathrm{T}} & 0 & \cdots & 0 & -r_{\mathrm{PET},1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ (\boldsymbol{p} - \boldsymbol{q}_{N_{PET}})^{\mathrm{T}} & 0 & \cdots & 0 & -r_{\mathrm{PET},N_{\mathrm{PET}}} \end{bmatrix}. \tag{14}$$

$$\boldsymbol{\theta}^0 = \left[\left(\boldsymbol{u}^0\right)^{\mathrm{T}}, L_{\mathrm{PCL},1}, \ldots, L_{\mathrm{PCL},N_{\mathrm{PCL}}}, L_{\mathrm{PET}}\right]^{\mathrm{T}}. \tag{15}$$

In (12), the number of equations is $N_{\mathrm{PCL}} + N_{\mathrm{PET}}$ and the number of unknowns is $N_{\mathrm{D}} + N_{\mathrm{PCL}} + 1$. The existing classic algorithms obtain an initial localization result by ignoring the constraint relationship between unknown variables. But when$N_{\mathrm{PET}} < N_{\mathrm{D}} + 1$, (12) is underdetermined and cannot be solved. Therefore, it is expected to use the constraint relationship between unknown variables to formulate the localization problem into a constrained optimization problem.

By replacing $r_{\mathrm{PCL},i}$ and $r_{\mathrm{PET},j}$ with $\tilde{r}_{\mathrm{PCL},i}$ and $\tilde{r}_{\mathrm{PET},j}$, (12) is converted into

$$\tilde{\boldsymbol{h}} - \tilde{\boldsymbol{G}}\boldsymbol{\theta} = \boldsymbol{\varepsilon} \tag{16}$$

where $\boldsymbol{\varepsilon}$ is the error term, $\tilde{\boldsymbol{h}} = \boldsymbol{h} + \Delta\boldsymbol{h}$, $\tilde{\boldsymbol{G}} = \boldsymbol{G} + \Delta\boldsymbol{G}$, and

$$\Delta\boldsymbol{h} = \begin{bmatrix} \boldsymbol{r}_{\mathrm{PCL}} \odot \boldsymbol{\delta}_{\mathrm{PCL}} + \frac{1}{2}\boldsymbol{\delta}_{\mathrm{PCL}} \odot \boldsymbol{\delta}_{\mathrm{PCL}} \\ \boldsymbol{r}_{\mathrm{PET}} \odot \boldsymbol{\delta}_{\mathrm{PET}} + \frac{1}{2}\boldsymbol{\delta}_{\mathrm{PET}} \odot \boldsymbol{\delta}_{\mathrm{PET}} \end{bmatrix} \approx \begin{bmatrix} \boldsymbol{r}_{\mathrm{PCL}} \odot \boldsymbol{\delta}_{\mathrm{PCL}} \\ \boldsymbol{r}_{\mathrm{PET}} \odot \boldsymbol{\delta}_{\mathrm{PET}} \end{bmatrix} \tag{17}$$

$$\Delta\boldsymbol{G} = \begin{bmatrix} \mathbf{O}_{N_{\mathrm{PCL}}\times N_{\mathrm{D}}} & \mathrm{diag}\left(\boldsymbol{\delta}_{\mathrm{PCL}}\right) & \mathbf{0}_{N_{\mathrm{PCL}}} \\ \mathbf{O}_{N_{\mathrm{PET}}\times N_{\mathrm{D}}} & \mathbf{O}_{N_{\mathrm{PET}}\times N_{\mathrm{PCL}}} & -\boldsymbol{\delta}_{\mathrm{PET}} \end{bmatrix}. \tag{18}$$

According to the least squares estimation and the constraint relationship between the elements in $\boldsymbol{\theta}$, the pseudolinear set of equations can be rewritten as a constrained optimization problem

$$\begin{aligned} &\min_{\boldsymbol{\theta}} E\left\{\|\boldsymbol{\varepsilon}\|^2\right\} \\ &\text{s.t.} \quad L_{\mathrm{PCL},i} = \|\boldsymbol{s}_i - \boldsymbol{u}\|\ i = 1, 2, \ldots, N_{\mathrm{PCL}} \\ &L_{\mathrm{PET}} = \|\boldsymbol{q} - \boldsymbol{u}\|. \end{aligned} \tag{19}$$

The objective function can be expanded as

$$\mathrm{E}\left\{\|\boldsymbol{\varepsilon}\|^2\right\} = \|\boldsymbol{h} - \boldsymbol{G}\boldsymbol{\theta}\|^2 + \boldsymbol{\theta}^T\boldsymbol{C}_G\boldsymbol{\theta} - 2\boldsymbol{\theta}^T\boldsymbol{\eta}_{Gh} + \boldsymbol{C}_h. \tag{20}$$

Since $\boldsymbol{C}_h$ is independent of the optimization variable $\boldsymbol{\theta}$, it can be ignored in subsequent discussions. $\boldsymbol{C}_G$ and $\boldsymbol{\eta}_{Gh}$ are

$$\begin{aligned}\boldsymbol{C}_G &= \mathrm{E}\left\{\Delta\boldsymbol{G}^{\mathrm{T}}\Delta\boldsymbol{G}\right\}\\ &= \mathrm{diag}\left(\boldsymbol{0}_{N_D}, \mathrm{diag}\left(\boldsymbol{Q}_{\mathrm{PCL}}\right), \mathrm{tr}\left(\boldsymbol{Q}_{\mathrm{PET}}\right)\right)\end{aligned} \tag{21}$$

$$\boldsymbol{\eta}_{Gh} = \mathrm{E}\left\{\Delta\boldsymbol{G}^{\mathrm{T}}\Delta\boldsymbol{h}\right\} = \begin{bmatrix}\boldsymbol{0}_{N_D}\\ \mathrm{diag}\left(\boldsymbol{Q}_{\mathrm{PCL}}\right)\odot\boldsymbol{r}_{\mathrm{PCL}}\\ -\mathrm{tr}\left(\mathrm{diag}\left(\boldsymbol{r}_{\mathrm{PET}}\right)\cdot\boldsymbol{Q}_{\mathrm{PET}}\right)\end{bmatrix}. \tag{22}$$

Let $\boldsymbol{A}$ be the square root of $\boldsymbol{G}^{\mathrm{T}}\boldsymbol{G}+\boldsymbol{C}_G$, and $\boldsymbol{b} = (\boldsymbol{A}^{\mathrm{T}})^{-1}(\boldsymbol{G}^{\mathrm{T}}\boldsymbol{h}+\boldsymbol{\eta}_{Gh})$. The problem in (19) can be transformed into

$$\begin{aligned}&\min_{\boldsymbol{\theta}} \|\boldsymbol{A}\boldsymbol{\theta}-\boldsymbol{b}\|^2\\ &\text{s.t.}\quad L_{\mathrm{PCL},i} = \|\boldsymbol{s}_i-\boldsymbol{u}\|\ i=1,2,\ldots,N_{\mathrm{PCL}}\\ &L_{\mathrm{PET}} = \|\boldsymbol{q}-\boldsymbol{u}\|.\end{aligned} \tag{23}$$

The optimization problem in (23) is QCQP, which is challenging to solve. Next, it will be transformed into a convex optimization problem through semidefinite relaxation. The objective function in (23) can be equivalently represented as

$$\|\boldsymbol{A}\boldsymbol{\theta}-\boldsymbol{b}\|^2 = \mathrm{tr}\left(\begin{bmatrix}\boldsymbol{\Theta} & \boldsymbol{\theta}\\ \boldsymbol{\theta}^{\mathrm{T}} & 1\end{bmatrix}\begin{bmatrix}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A} & -\boldsymbol{A}^{\mathrm{T}}\boldsymbol{b}\\ -\boldsymbol{b}^{\mathrm{T}}\boldsymbol{A} & \boldsymbol{b}^{\mathrm{T}}\boldsymbol{b}\end{bmatrix}\right) \tag{24}$$

where $\boldsymbol{\Theta} = \boldsymbol{\theta}\boldsymbol{\theta}^{\mathrm{T}}$. The constraints can be expressed as

$$\begin{aligned}&[\Theta]_{N_{\mathrm{D}}+i,N_{\mathrm{D}}+i} = \mathrm{tr}\left([\Theta]_{1:N_{\mathrm{D}},1:N_{\mathrm{D}}}\right)\\ &\quad-2\boldsymbol{s}_i{}^{\mathrm{T}}[\theta]_{1:N_{\mathrm{D}}}+\boldsymbol{s}_i{}^{\mathrm{T}}\boldsymbol{s}_i,\ i=1,2,\ldots,N_{\mathrm{PCL}}\end{aligned} \tag{25}$$

$$\begin{aligned}&[\Theta]_{N_{\mathrm{D}}+N_{\mathrm{PCL}}+1,N_{\mathrm{D}}+N_{\mathrm{PCL}}+1} = \mathrm{tr}\left([\Theta]_{1:N_{\mathrm{D}},1:N_{\mathrm{D}}}\right)\\ &\quad-2\boldsymbol{p}^{\mathrm{T}}[\theta]_{1:N_{\mathrm{D}}}+\boldsymbol{p}^{\mathrm{T}}\boldsymbol{p}.\end{aligned} \tag{26}$$

The additional constraint $\boldsymbol{\Theta} = \boldsymbol{\theta}\boldsymbol{\theta}^{\mathrm{T}}$ can be relaxed to $\boldsymbol{\Theta} \succcurlyeq \boldsymbol{\theta}\boldsymbol{\theta}^{\mathrm{T}}$ and $\mathrm{rank}(\boldsymbol{\Theta}) = 1$. By disregarding the nonconvex constraint of the matrix rank being 1, we can obtain the final optimization problem

$$\begin{aligned}&\min_{\boldsymbol{\theta},\Theta} \|\boldsymbol{A}\boldsymbol{\theta}-\boldsymbol{b}\|^2\\ &\text{s.t.}\quad (25)-(26)\\ &\begin{bmatrix}\Theta & \theta\\ \theta^{\mathrm{T}} & 1\end{bmatrix} \succcurlyeq 0.\end{aligned} \tag{27}$$

The optimization problem can be solved using the CVX toolbox. It is worth noting that appropriate normalization of parameters is required. Subsequently, the localization accuracy can further be improved by considering the error distribution of measurements. Following the approach in [21], the localization result of (27) is represented as the true value plus the error

$$\hat{\boldsymbol{u}} = \boldsymbol{u}^0+\Delta\boldsymbol{u}. \tag{28}$$

The auxiliary variables $L_{\mathrm{PCL},i}$ and L_{PET} are subjected to a first-order Taylor expansion around $\hat{\boldsymbol{u}}$, with higher order terms being disregarded

$$\begin{aligned}L_{\mathrm{PCL},i} &= \left\|\boldsymbol{s}_i-\boldsymbol{u}^0\right\| = \|\boldsymbol{s}_i-\hat{\boldsymbol{u}}+\Delta\boldsymbol{u}\|\\ &\approx \|\boldsymbol{s}_i-\hat{\boldsymbol{u}}\|+\frac{(\boldsymbol{s}_i-\hat{\boldsymbol{u}})^{\mathrm{T}}}{\|\boldsymbol{s}_i-\hat{\boldsymbol{u}}\|}\Delta\boldsymbol{u},\ i=1,2,\ldots,N_{\mathrm{PCL}}\end{aligned} \tag{29}$$

$$\begin{aligned}L_{\mathrm{PET}} &= \left\|\boldsymbol{p}-\boldsymbol{u}^0\right\| = \|\boldsymbol{p}-\hat{\boldsymbol{u}}+\Delta\boldsymbol{u}\|\\ &\approx \|\boldsymbol{p}-\hat{\boldsymbol{u}}\|+\frac{(\boldsymbol{p}-\hat{\boldsymbol{u}})^{\mathrm{T}}}{\|\boldsymbol{p}-\hat{\boldsymbol{u}}\|}\Delta\boldsymbol{u}.\end{aligned} \tag{30}$$

Substituting (28)–(30) into (10) and (11) yields linear equations for the localization error

$$\begin{aligned}&\frac{1}{2}\left(\tilde{r}_{\mathrm{PCL},i}^2+\|\boldsymbol{s}_i\|^2-\|\boldsymbol{t}_i\|^2\right)-(\boldsymbol{s}_i-\boldsymbol{t}_i)^{\mathrm{T}}\hat{\boldsymbol{u}}\\ &-\tilde{r}_{\mathrm{PCL},i}\|\boldsymbol{s}_i-\hat{\boldsymbol{u}}\|-\left[\tilde{r}_{\mathrm{PCL},i}\frac{(\boldsymbol{s}_i-\hat{\boldsymbol{u}})^{\mathrm{T}}}{\|\boldsymbol{s}_i-\hat{\boldsymbol{u}}\|}-(\boldsymbol{s}_i-\boldsymbol{t}_i)^{\mathrm{T}}\right]\Delta\boldsymbol{u}\\ &\approx\left(\tilde{r}_{\mathrm{PCL},i}-\|\boldsymbol{s}_i-\hat{\boldsymbol{u}}\|\right)\delta_{\mathrm{PCL},i}\end{aligned} \tag{31}$$

$$\begin{aligned}&\frac{1}{2}\left(\tilde{r}_{\mathrm{PET},j}^2+\|\boldsymbol{p}\|^2-\left\|\boldsymbol{q}_j\right\|^2\right)-\left(\boldsymbol{p}-\boldsymbol{q}_j\right)^{\mathrm{T}}\hat{\boldsymbol{u}}\\ &+\tilde{r}_{\mathrm{PET},j}\|\boldsymbol{p}-\hat{\boldsymbol{u}}\|+\left[\tilde{r}_{\mathrm{PET},j}\frac{(\boldsymbol{p}-\hat{\boldsymbol{u}})^{\mathrm{T}}}{\|\boldsymbol{p}-\hat{\boldsymbol{u}}\|}+\left(\boldsymbol{p}-\boldsymbol{q}_j\right)^{\mathrm{T}}\right]\Delta\boldsymbol{u}\\ &\approx\left(\tilde{r}_{\mathrm{PET},j}+\|\boldsymbol{p}-\hat{\boldsymbol{u}}\|\right)\delta_{\mathrm{PET},j}.\end{aligned} \tag{32}$$

Stacking (31) and (32) yields

$$\boldsymbol{d}-\boldsymbol{F}\Delta\boldsymbol{u} = \boldsymbol{C}\begin{bmatrix}\delta_{\mathrm{PCL}}\\ \delta_{\mathrm{PET}}\end{bmatrix} \tag{33}$$

where $\boldsymbol{d} = [\boldsymbol{d}_1^{\mathrm{T}}, \boldsymbol{d}_2^{\mathrm{T}}]^{\mathrm{T}}$, $\boldsymbol{F} = [\boldsymbol{F}_1^{\mathrm{T}}, \boldsymbol{F}_2^{\mathrm{T}}]^{\mathrm{T}}$

$$\begin{aligned}\left[\boldsymbol{d}_1\right]_i &= \frac{1}{2}\left(\tilde{r}_{\mathrm{PCL},i}^2+\|\boldsymbol{s}_i\|^2-\|\boldsymbol{t}_i\|^2\right)-(\boldsymbol{s}_i-\boldsymbol{t}_i)^{\mathrm{T}}\hat{\boldsymbol{u}}\\ &-\tilde{r}_{\mathrm{PCL},i}\|\boldsymbol{s}_i-\hat{\boldsymbol{u}}\|\end{aligned} \tag{34}$$

$$\begin{aligned}\left[\boldsymbol{d}_2\right]_j &= \frac{1}{2}\left(\tilde{r}_{\mathrm{PET},j}^2+\|\boldsymbol{p}\|^2-\left\|\boldsymbol{q}_j\right\|^2\right)-\left(\boldsymbol{p}-\boldsymbol{q}_j\right)^{\mathrm{T}}\hat{\boldsymbol{u}}\\ &+\tilde{r}_{\mathrm{PET},j}\|\boldsymbol{p}-\hat{\boldsymbol{u}}\|\end{aligned} \tag{35}$$

$$[\boldsymbol{F}_1]_{i,:} = \tilde{r}_{\mathrm{PCL},1}(\boldsymbol{s}_i-\hat{\boldsymbol{u}})^{\mathrm{T}}/\|\boldsymbol{s}_i-\hat{\boldsymbol{u}}\|-(\boldsymbol{s}_i-\boldsymbol{t}_i)^{\mathrm{T}} \tag{36}$$

$$[\boldsymbol{F}_2]_{j,:} = -\tilde{r}_{\mathrm{PET},j}(\boldsymbol{p}-\hat{\boldsymbol{u}})^{\mathrm{T}}/\|\boldsymbol{p}-\hat{\boldsymbol{u}}\|-\left(\boldsymbol{p}-\boldsymbol{q}_j\right)^{\mathrm{T}} \tag{37}$$

$$\boldsymbol{C} = \mathrm{diag}\left(\left[\tilde{\boldsymbol{r}}_{\mathrm{PCL}}^{\mathrm{T}}, \tilde{\boldsymbol{r}}_{\mathrm{PET}}^{\mathrm{T}}\right]^{\mathrm{T}}\right)+\mathrm{diag}\left(\left[\boldsymbol{c}_{\mathrm{PCL}}^{\mathrm{T}}, \boldsymbol{c}_{\mathrm{PET}}^{\mathrm{T}}\right]^{\mathrm{T}}\right) \tag{38}$$

$$\boldsymbol{c}_{\mathrm{PCL}} = -\left[\|\boldsymbol{s}_1-\hat{\boldsymbol{u}}\|,\ldots,\left\|\boldsymbol{s}_{N_{\mathrm{PCL}}}-\hat{\boldsymbol{u}}\right\|\right]^{\mathrm{T}} \tag{39}$$

$$\boldsymbol{c}_{\mathrm{PET}} = \left[\|\boldsymbol{p}-\hat{\boldsymbol{u}}\|,\ldots,\|\boldsymbol{p}-\hat{\boldsymbol{u}}\|\right]^{\mathrm{T}}. \tag{40}$$

The weighted least squares solution of (33) is

$$\Delta\hat{\boldsymbol{u}} = \left(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{W}^{-1}\boldsymbol{F}\right)^{-1}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{W}^{-1}\boldsymbol{d} \tag{41}$$

where $\boldsymbol{W} = \boldsymbol{C}\boldsymbol{Q}\boldsymbol{C}^{\mathrm{T}}$ and $\boldsymbol{Q} =$blkdiag$(\boldsymbol{Q}_{\mathrm{PCL}}, \boldsymbol{Q}_{\mathrm{PET}})$. Thus, the final target localization result is

$$\bar{\boldsymbol{u}} = \hat{\boldsymbol{u}}-\Delta\hat{\boldsymbol{u}}. \tag{42}$$

This algorithm does not require prior information about the target position, and the localization accuracy can approach the CRLB even under significant measurement errors. In the following simulations, the results obtained from (27) are denoted as the SDP algorithm, while the results obtained from (42) are denoted as the "SDP+Improvement" algorithm.

B. Newton–Homotopy-Based Localization Algorithm

This section discusses the localization problem under the condition of known prior information about the target position. When the target's position at the previous moment is known or when a specific PCL receiver in the network can measure the azimuth angle of the target (PCL system angle measurement errors are often significant), a rough initial value of the target position can be obtained. Therefore, it is meaningful to discuss the problem of target localization in this scenario.

Suppose that in a two-dimensional scenario, the ith PCL receiver can measure the target's azimuth angle $\tilde{\alpha}_i$. Then, the initial position estimation $\hat{\boldsymbol{u}}$ can be obtained by solving the triangle, that is

$$\hat{\boldsymbol{u}} = \frac{\|\boldsymbol{s}_i - \boldsymbol{t}_i\|^2 - \tilde{r}_{\mathrm{PCL},i}^2}{2\left[\|\boldsymbol{s}_i - \boldsymbol{t}_i\| \cos\left(\tilde{\alpha}_i - \alpha_{t_i s_i}\right) - r_{\mathrm{PCL},i}\right]} \begin{bmatrix} \sin\tilde{\alpha}_i \\ \cos\tilde{\alpha}_i \end{bmatrix} + \boldsymbol{s}_i \tag{43}$$

where $\alpha_{t_i s_i}$ represents the azimuth angle of $\boldsymbol{t}_i$ relative to $\boldsymbol{s}_i$. For the target localization problem of PCL-PET hybrid heterogeneous, the likelihood function can be expressed as

$$l\left(\tilde{\boldsymbol{r}}; \boldsymbol{u}\right) = \frac{1}{\sqrt{\det\left(2\pi \boldsymbol{Q}\right)}} \cdot \exp\left\{-\frac{1}{2}\left[\boldsymbol{g}\left(\boldsymbol{u}\right) - \tilde{\boldsymbol{r}}\right]^{\mathrm{T}} \boldsymbol{Q}^{-1}\left[\boldsymbol{g}\left(\boldsymbol{u}\right) - \tilde{\boldsymbol{r}}\right]\right\} \tag{44}$$

where $\tilde{\boldsymbol{r}} = [\tilde{\boldsymbol{r}}_{\mathrm{PCL}}^{\mathrm{T}}, \tilde{\boldsymbol{r}}_{\mathrm{PET}}^{\mathrm{T}}]^{\mathrm{T}}$, $\boldsymbol{g}(\boldsymbol{u}) = [\boldsymbol{g}_{\mathrm{PCL}}^{\mathrm{T}}, \boldsymbol{g}_{\mathrm{PET}}^{\mathrm{T}}]^{\mathrm{T}}$

$$\tilde{\boldsymbol{r}}_{\mathrm{PCL}} = [\tilde{r}_{\mathrm{PCL},1}, \tilde{r}_{\mathrm{PCL},2}, \ldots, \tilde{r}_{\mathrm{PCL},N_{\mathrm{PCL}}}]^{\mathrm{T}}$$

$$\tilde{\boldsymbol{r}}_{\mathrm{PET}} = [\tilde{r}_{\mathrm{PET},1}, \tilde{r}_{\mathrm{PET},2}, \ldots, \tilde{r}_{\mathrm{PET},N_{\mathrm{PET}}}]^{\mathrm{T}}$$

$$\boldsymbol{g}_{\mathrm{PCL}}(\boldsymbol{u}) = \begin{bmatrix} \|\boldsymbol{u} - \boldsymbol{s}_1\| + \|\boldsymbol{u} - \boldsymbol{t}_1\| \\ \vdots \\ \|\boldsymbol{u} - \boldsymbol{s}_{N_{\mathrm{PCL}}}\| + \|\boldsymbol{u} - \boldsymbol{t}_{N_{\mathrm{PCL}}}\| \end{bmatrix}$$

$$\boldsymbol{g}_{\mathrm{PET}}(\boldsymbol{u}) = \begin{bmatrix} \|\boldsymbol{u} - \boldsymbol{q}_1\| - \|\boldsymbol{u} - \boldsymbol{p}\| \\ \vdots \\ \|\boldsymbol{u} - \boldsymbol{q}_{N_{\mathrm{PET}}}\| - \|\boldsymbol{u} - \boldsymbol{p}\| \end{bmatrix}.$$

The corresponding logarithmic likelihood function is

$$\ln l\left(\tilde{\boldsymbol{r}}; \boldsymbol{u}\right) = \mathrm{const} - \frac{1}{2}\left[\boldsymbol{g}\left(\boldsymbol{u}\right) - \tilde{\boldsymbol{r}}\right]^{\mathrm{T}} \boldsymbol{Q}^{-1}\left[\boldsymbol{g}\left(\boldsymbol{u}\right) - \tilde{\boldsymbol{r}}\right]. \tag{45}$$

In (45), "const" represents a constant independent of both $\tilde{\boldsymbol{r}}$ and $\boldsymbol{u}$. Let

$$f\left(\boldsymbol{u}\right) = \left[\boldsymbol{g}\left(\boldsymbol{u}\right) - \tilde{\boldsymbol{r}}\right]^{\mathrm{T}} \boldsymbol{Q}^{-1}\left[\boldsymbol{g}\left(\boldsymbol{u}\right) - \tilde{\boldsymbol{r}}\right] \tag{46}$$

under the maximum likelihood criterion, the target localization problem can be described as

$$\hat{\boldsymbol{u}}_{\mathrm{ML}} = \underset{\boldsymbol{u}}{\operatorname{argmin}}\, f\left(\boldsymbol{u}\right). \tag{47}$$

Equation (47) is an unconstrained nonlinear minimization problem, which typically has no closed-form solution. The Newton–Raphson method can solve this problem iteratively. Because it utilizes the second-order derivative of the likelihood function, it can achieve a fast convergence speed. However, the Newton–Raphson method requires high accuracy in iterative initial values and is prone to divergence when the initial error is large. This article proposes an algorithm based on the Newton–Homotopy method, which can achieve convergence results with larger initial errors. To the best of our knowledge, there are currently no reports on using the Newton–Homotopy method to solve the target localization problem.

The first-order partial derivative of $f(\boldsymbol{u})$ with respect to the target position $\boldsymbol{u}$ is defined as

$$\nabla f\left(\boldsymbol{u}\right) \triangleq \frac{\partial f\left(\boldsymbol{u}\right)}{\partial \boldsymbol{u}} = 2\frac{\partial \boldsymbol{g}^{\mathrm{T}}\left(\boldsymbol{u}\right)}{\partial \boldsymbol{u}} \boldsymbol{Q}^{-1}\left[\boldsymbol{g}\left(\boldsymbol{u}\right) - \tilde{\boldsymbol{r}}\right]. \tag{48}$$

The localization problem is transformed into solving $\nabla f(\boldsymbol{u}) = \boldsymbol{0}_{N_{\mathrm{D}}}$. Assuming that $\boldsymbol{f}_0(\boldsymbol{u})$ is a known function with a known zero point $\boldsymbol{u}^*$, i.e., $\boldsymbol{f}_0(\boldsymbol{u}^*) = \boldsymbol{0}_{N_{\mathrm{D}}}$. The homotopy function can be constructed as

$$\boldsymbol{h}\left(\boldsymbol{u}, s\right) = s\nabla f\left(\boldsymbol{u}\right) + \left(1 - s\right)\boldsymbol{f}_0\left(\boldsymbol{u}\right), s \in [0, 1]. \tag{49}$$

Obviously, when $s = 0$, $\boldsymbol{h}(\boldsymbol{u}, 0) = \boldsymbol{0}_{N_{\mathrm{D}}}$ is a system of equations with known solutions. When $s = 1$, $\boldsymbol{h}(\boldsymbol{u}, 1) = \boldsymbol{0}_{N_{\mathrm{D}}}$ is the original problem. This reflects the idea of the homotopy method: constructing a continuous path of functions that gradually transforms the original complex problem into a series of simpler problems. Along this path, the solutions of the equations evolve as the path parameter changes. By progressively adjusting the path parameter, one can transition from a known simple problem to the original problem and ultimately obtain the solution.

Let

$$\boldsymbol{f}_0\left(\boldsymbol{u}\right) = \nabla f\left(\boldsymbol{u}\right) - \nabla f\left(\hat{\boldsymbol{u}}\right) \tag{50}$$

then the zero point of $\boldsymbol{f}_0(\boldsymbol{u})$ is $\hat{\boldsymbol{u}}$. Substituting (50) into (49), the homotopy function can be simplified as

$$\boldsymbol{h}\left(\boldsymbol{u}, s\right) = \nabla f\left(\boldsymbol{u}\right) + \left(s - 1\right)\nabla f\left(\hat{\boldsymbol{u}}\right), s \in [0, 1]. \tag{51}$$

Considering that the solution of $\boldsymbol{h}(\boldsymbol{u}, s) = \boldsymbol{0}_{N_{\mathrm{D}}}$ depends on s, it is denoted as $\hat{\boldsymbol{u}}(s)$. Discretize the value interval of s to $0 = s_0 < s_1 < \ldots < s_n = 1$ and solve a series of nonlinear equations

$$\boldsymbol{h}\left(\boldsymbol{u}, s_i\right) = \boldsymbol{0}_{N_{\mathrm{D}}}. \tag{52}$$

The initial value of each iteration is $\hat{\boldsymbol{u}}(s_{i-1})$, which is the result of the previous iteration. Algorithm 1 provides detailed steps for the Newton–Homotopy-based target localization algorithm. The input parameters of the algorithm include the initial position estimation $\hat{\boldsymbol{u}}$, maximum number of iterations k_{MAX}, and iteration termination condition k_{TOL}. The definition of $\boldsymbol{H}(\boldsymbol{u}, s_i)$ in step 6 is

$$\boldsymbol{H}\left(\boldsymbol{u}, s_i\right) \triangleq \frac{\partial \boldsymbol{h}\left(\boldsymbol{u}, s_i\right)}{\partial \boldsymbol{u}^{\mathrm{T}}}. \tag{53}$$

In the Appendix, we provide a detailed expression for $\boldsymbol{H}(\boldsymbol{u}, s_i)$ and $\boldsymbol{h}(\boldsymbol{u}, s_i)$ in a two-dimensional localization scenario. The final output of the Newton–Homotopy algorithm is $\hat{\boldsymbol{u}}_{\mathrm{ML}}$.

Algorithm 1: Newton–Homotopy-based target localization

Input: $\hat{\boldsymbol{u}}, k_{\mathrm{MAX}}, k_{\mathrm{TOL}}$

Interval discretization
$0 = s_0 < s_1 < \ldots < s_n = 1$
$\hat{\boldsymbol{u}}(s_0) = \hat{\boldsymbol{u}}$
for $i = 1 : n$ **do**
  $\hat{\boldsymbol{u}}(s_i) = \hat{\boldsymbol{u}}(s_{i-1}),\ j = 1$
  while $j \leq k_{\mathrm{MAX}}$ **do**
    $\hat{\boldsymbol{u}}(s_i) = \hat{\boldsymbol{u}}(s_i)$
    $- \boldsymbol{H}(\hat{\boldsymbol{u}}(s_i), s_i)^{-1}\boldsymbol{h}(\hat{\boldsymbol{u}}(s_i), s_i)$
    $j = j + 1$
    if $\|\boldsymbol{h}(\boldsymbol{u}, s_i)\| < k_{\mathrm{TOL}}$ **then**
      break
    end if
  end while
end for
$\hat{\boldsymbol{u}}_{\mathrm{ML}} = \hat{\boldsymbol{u}}(s_n)$

Output: $\hat{\boldsymbol{u}}_{\mathrm{ML}}$

C. Measurement Association Algorithm

For the correct measurement combinations, the two localization algorithms mentioned above can obtain the positions of the targets. However, the localization algorithms will produce ghosts for the incorrect measurement combinations. Therefore, making additional association decisions on the localization results is necessary. The results that pass the decision will be recognized as true targets.

Suppose that $\tilde{\boldsymbol{r}}' = [\tilde{r}'_{\mathrm{PCL},1}, \ldots, \tilde{r}'_{\mathrm{PCL},N_{\mathrm{PCL}}}, \tilde{r}'_{\mathrm{PET},1}, \ldots, \tilde{r}'_{\mathrm{PET},N_{\mathrm{PET}}}]^{\mathrm{T}}$ is a potential measurement combination. The maximum likelihood estimate $\hat{\boldsymbol{u}}_{\mathrm{ML}}$ of the combination has been obtained through one of the algorithms mentioned above, that is

$$\hat{\boldsymbol{u}}_{\mathrm{ML}} = \underset{\boldsymbol{u}}{\operatorname{argmin}} \left[\tilde{\boldsymbol{r}}' - \boldsymbol{g}(\boldsymbol{u})\right]^{\mathrm{T}} \boldsymbol{Q}^{-1} \left[\tilde{\boldsymbol{r}}' - \boldsymbol{g}(\boldsymbol{u})\right]. \tag{54}$$

Subsequently, we will prove that when $\tilde{\boldsymbol{r}}'$ is the correct combination, the test statistic Υ approximately follows a chi-square distribution with $N_{\mathrm{PCL}} + N_{\mathrm{PET}} - N_{\mathrm{D}}$ degrees of freedom. The definition of Υ is

$$\Upsilon \triangleq \left[\tilde{\boldsymbol{r}}' - \boldsymbol{g}\left(\hat{\boldsymbol{u}}_{\mathrm{ML}}\right)\right]^{\mathrm{T}} \boldsymbol{Q}^{-1} \left[\tilde{\boldsymbol{r}}' - \boldsymbol{g}\left(\hat{\boldsymbol{u}}_{\mathrm{ML}}\right)\right]. \tag{55}$$

When $\tilde{\boldsymbol{r}}'$ is the correct combination, $\hat{\boldsymbol{u}}_{\mathrm{ML}}$ is close to the true value $\boldsymbol{u}^0$, denoted as $\hat{\boldsymbol{u}}_{\mathrm{ML}} = \boldsymbol{u}^0 + \Delta\boldsymbol{u}$. Therefore, the measurement function can be approximated as

$$\boldsymbol{g}\left(\hat{\boldsymbol{u}}_{\mathrm{ML}}\right) \approx \boldsymbol{g}\left(\boldsymbol{u}^0\right) + \boldsymbol{G}_{\boldsymbol{u}^0}\Delta\boldsymbol{u} \tag{56}$$

where $\boldsymbol{G}_{\boldsymbol{u}^0} = \frac{\partial \boldsymbol{g}(\boldsymbol{u})}{\partial \boldsymbol{u}^{\mathrm{T}}}\Big|_{\boldsymbol{u}=\boldsymbol{u}^0}$. Substituting (56) into (54) yields an approximate estimate of the localization error as

$$\Delta\hat{\boldsymbol{u}} \approx \underset{\Delta\boldsymbol{u}}{\operatorname{argmin}} \left[\boldsymbol{\delta}' - \boldsymbol{G}_{\boldsymbol{u}^0}\Delta\boldsymbol{u}\right]^{\mathrm{T}} \boldsymbol{Q}^{-1} \left[\boldsymbol{\delta}' - \boldsymbol{G}_{\boldsymbol{u}^0}\Delta\boldsymbol{u}\right]. \tag{57}$$

$\boldsymbol{\delta}'$ represents the measurement error vector of $\tilde{\boldsymbol{r}}'$. Due to $\boldsymbol{Q}$ being the covariance matrix, possessing positive-definite characteristics, $\boldsymbol{Q}^{-1}$ can thus be decomposed as

$$\boldsymbol{Q}^{-1} = \boldsymbol{D}^{\mathrm{T}}\boldsymbol{D}. \tag{58}$$

Equation (57) can be transformed into

$$\begin{aligned} \Delta\hat{\boldsymbol{u}} &\approx \underset{\Delta\boldsymbol{u}}{\operatorname{argmin}} \left[\boldsymbol{D}\boldsymbol{\delta}' - \boldsymbol{D}\boldsymbol{G}_{\boldsymbol{u}^0}\Delta\boldsymbol{u}\right]^{\mathrm{T}} \left[\boldsymbol{D}\boldsymbol{\delta}' - \boldsymbol{D}\boldsymbol{G}_{\boldsymbol{u}^0}\Delta\boldsymbol{u}\right] \\ &= \underset{\Delta\boldsymbol{u}}{\operatorname{argmin}} \left[\bar{\boldsymbol{\delta}} - \bar{\boldsymbol{G}}\Delta\boldsymbol{u}\right]^{\mathrm{T}} \left[\bar{\boldsymbol{\delta}} - \bar{\boldsymbol{G}}\Delta\boldsymbol{u}\right] \\ &= \left(\bar{\boldsymbol{G}}^{\mathrm{T}}\bar{\boldsymbol{G}}\right)^{-1}\bar{\boldsymbol{G}}^{\mathrm{T}}\bar{\boldsymbol{\delta}} \end{aligned} \tag{59}$$

where $\bar{\boldsymbol{G}} = \boldsymbol{D}\boldsymbol{G}_{\boldsymbol{u}^0}$ and $\bar{\boldsymbol{\delta}} = \boldsymbol{D}\boldsymbol{\delta}'$. In addition, it can be proved that $\bar{\boldsymbol{\delta}} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_{N_{\mathrm{PCL}}+N_{\mathrm{PET}}})$. The residual $\Delta\bar{\boldsymbol{x}}$ can be expressed as

$$\Delta\bar{\boldsymbol{x}} = \bar{\boldsymbol{\delta}} - \bar{\boldsymbol{G}}\Delta\hat{\boldsymbol{u}} \approx \left[\boldsymbol{I}_{N_{\mathrm{PCL}}+N_{\mathrm{PET}}} - \bar{\boldsymbol{G}}\left(\bar{\boldsymbol{G}}^{\mathrm{T}}\bar{\boldsymbol{G}}\right)^{-1}\bar{\boldsymbol{G}}^{\mathrm{T}}\right]\bar{\boldsymbol{\delta}}. \tag{60}$$

The residual $\Delta\bar{\boldsymbol{x}}$ is the orthogonal projection onto the column space from $\bar{\boldsymbol{\delta}}$ to $\bar{\boldsymbol{G}}$. Based on the linearity invariance of normal variables and the idempotence of orthogonal projection matrices, it can be inferred that the residual $\Delta\bar{\boldsymbol{x}}$ still follows a zero-mean Gaussian distribution, with the covariance matrix given by

$$\operatorname{Cov}\left(\Delta\bar{\boldsymbol{x}}\right) = \mathrm{E}\left\{\Delta\bar{\boldsymbol{x}}\Delta\bar{\boldsymbol{x}}^{\mathrm{T}}\right\} \approx \boldsymbol{I}_{N_{\mathrm{PCL}}+N_{\mathrm{PET}}} - \bar{\boldsymbol{G}}\left(\bar{\boldsymbol{G}}^{\mathrm{T}}\bar{\boldsymbol{G}}\right)^{-1}\bar{\boldsymbol{G}}^{\mathrm{T}}. \tag{61}$$

Let $\boldsymbol{U} = [\boldsymbol{U}_1, \boldsymbol{U}_2]$ be an $(N_{\mathrm{PCL}} + N_{\mathrm{PET}}) \times (N_{\mathrm{PCL}} + N_{\mathrm{PET}})$ orthogonal matrix, and let the column space of $\boldsymbol{U}_1$ be identical to that of $\bar{\boldsymbol{G}}$. According to the definition of orthogonal projection, we have

$$\boldsymbol{I}_{N_{\mathrm{PCL}}+N_{\mathrm{PET}}} - \bar{\boldsymbol{G}}\left(\bar{\boldsymbol{G}}^{\mathrm{T}}\bar{\boldsymbol{G}}\right)^{-1}\bar{\boldsymbol{G}}^{\mathrm{T}} = \boldsymbol{U}_2{\boldsymbol{U}_2}^{\mathrm{T}}. \tag{62}$$

Taking $\Delta\bar{\boldsymbol{x}}' = {\boldsymbol{U}_2}^{\mathrm{T}}\Delta\bar{\boldsymbol{x}}$, then $\Delta\bar{\boldsymbol{x}}'$ follows a zero-mean Gaussian distribution in $N_{\mathrm{PCL}} + N_{\mathrm{PET}} - N_{\mathrm{D}}$ dimensions, with its covariance matrix being

$$\begin{aligned} &\operatorname{Cov}\left(\Delta\bar{\boldsymbol{x}}'\right) = E\left\{\Delta\bar{\boldsymbol{x}}'\Delta\bar{\boldsymbol{x}}'^{\mathrm{T}}\right\} \\ &\approx {\boldsymbol{U}_2}^{\mathrm{T}}\left[\boldsymbol{I}_{N_{\mathrm{PCL}}+N_{\mathrm{PET}}} - \bar{\boldsymbol{G}}\left(\bar{\boldsymbol{G}}^{\mathrm{T}}\bar{\boldsymbol{G}}\right)^{-1}\bar{\boldsymbol{G}}^{\mathrm{T}}\right]\boldsymbol{U}_2 \\ &= {\boldsymbol{U}_2}^{\mathrm{T}}\boldsymbol{U}_2{\boldsymbol{U}_2}^{\mathrm{T}}\boldsymbol{U}_2 = \boldsymbol{I}_{N_{\mathrm{PCL}}+N_{\mathrm{PET}}-N_{\mathrm{D}}}. \end{aligned} \tag{63}$$

According to the definition of the chi-square distribution, it can be inferred that $\Delta\bar{\boldsymbol{x}}'^{\mathrm{T}}\Delta\bar{\boldsymbol{x}}'$ approximately follows a chi-square distribution with $N_{\mathrm{PCL}} + N_{\mathrm{PET}} - N_{\mathrm{D}}$ degrees of freedom, denoted as $\Delta\bar{\boldsymbol{x}}'^{\mathrm{T}}\Delta\bar{\boldsymbol{x}}' \dot{\sim} \chi^2(N_{\mathrm{PCL}} + N_{\mathrm{PET}} - N_{\mathrm{D}})$. In addition, since

$$\Delta\bar{\boldsymbol{x}}^{\mathrm{T}}\Delta\bar{\boldsymbol{x}} = \Delta\bar{\boldsymbol{x}}^{\mathrm{T}}\boldsymbol{U}\boldsymbol{U}^{\mathrm{T}}\Delta\bar{\boldsymbol{x}} = \Delta\bar{\boldsymbol{x}}^{\mathrm{T}}\boldsymbol{U}_1{\boldsymbol{U}_1}^{\mathrm{T}}\Delta\bar{\boldsymbol{x}} + \Delta\bar{\boldsymbol{x}}^{\mathrm{T}}\boldsymbol{U}_2{\boldsymbol{U}_2}^{\mathrm{T}}\Delta\bar{\boldsymbol{x}} \tag{64}$$

and

$${\boldsymbol{U}_1}^{\mathrm{T}}\Delta\bar{\boldsymbol{x}} \approx {\boldsymbol{U}_1}^{\mathrm{T}}\boldsymbol{U}_2{\boldsymbol{U}_2}^{\mathrm{T}}\bar{\boldsymbol{\delta}} = \boldsymbol{0}_{N_D} \tag{65}$$

we have

$$\Delta\bar{\boldsymbol{x}}^{\mathrm{T}}\Delta\bar{\boldsymbol{x}} \approx \Delta\bar{\boldsymbol{x}}^{\mathrm{T}}\boldsymbol{U}_2{\boldsymbol{U}_2}^{\mathrm{T}}\Delta\bar{\boldsymbol{x}} = \Delta\bar{\boldsymbol{x}}'^{\mathrm{T}}\Delta\bar{\boldsymbol{x}}'. \tag{66}$$

Therefore, $\Delta\bar{\boldsymbol{x}}^{\mathrm{T}}\Delta\bar{\boldsymbol{x}}$ also approximately follows a chi-square distribution with $N_{\mathrm{PCL}} + N_{\mathrm{PET}} - N_{\mathrm{D}}$ degrees of freedom. Substituting (56) into (55) yields

$$\begin{aligned} \Upsilon &\approx \left[\boldsymbol{\delta}' - \boldsymbol{G}_{\boldsymbol{u}^0}\Delta\boldsymbol{u}\right]^{\mathrm{T}}\boldsymbol{Q}^{-1}\left[\boldsymbol{\delta}' - \boldsymbol{G}_{\boldsymbol{u}^0}\Delta\boldsymbol{u}\right] \\ &= \left[\bar{\boldsymbol{\delta}} - \bar{\boldsymbol{G}}\Delta\boldsymbol{u}\right]^{\mathrm{T}}\left[\bar{\boldsymbol{\delta}} - \bar{\boldsymbol{G}}\Delta\boldsymbol{u}\right] \end{aligned}$$

$$= \Delta\bar{\boldsymbol{x}}^{\mathrm{T}}\Delta\bar{\boldsymbol{x}} \tag{67}$$

hence $\Upsilon \dot{\sim} \chi^2(N_{\mathrm{PCL}} + N_{\mathrm{PET}} - N_{\mathrm{D}})$. All the approximations made during the derivation are based on the linearization of (56), which is acceptable when $\hat{\boldsymbol{u}}_{\mathrm{ML}}$ is close to the true value $\boldsymbol{u}^0$. The criterion for decision-making on the measurement association hypothesis is

$$\Upsilon \underset{H_1}{\overset{H_0}{\gtrless}} \gamma \tag{68}$$

where γ is the threshold for measurement association decision, and according to the required correct correlation probability P_{A} and the degree of freedom of the chi-square distribution, the threshold γ can be determined by querying the chi-square distribution table. When the test statistic Υ is greater than γ, it indicates acceptance of the following: “The measurement association hypothesis is incorrect”; when the test statistic Υ is less than γ, it indicates acceptance of the following: “The measurement association hypothesis is correct.”

IV. SIMULATION RESULTS

The target detection scenario set in this section is in a two-dimensional space. It is worth noting that the algorithms proposed in this article are also applicable to target localization problems and measurement association problems in three-dimensional space.

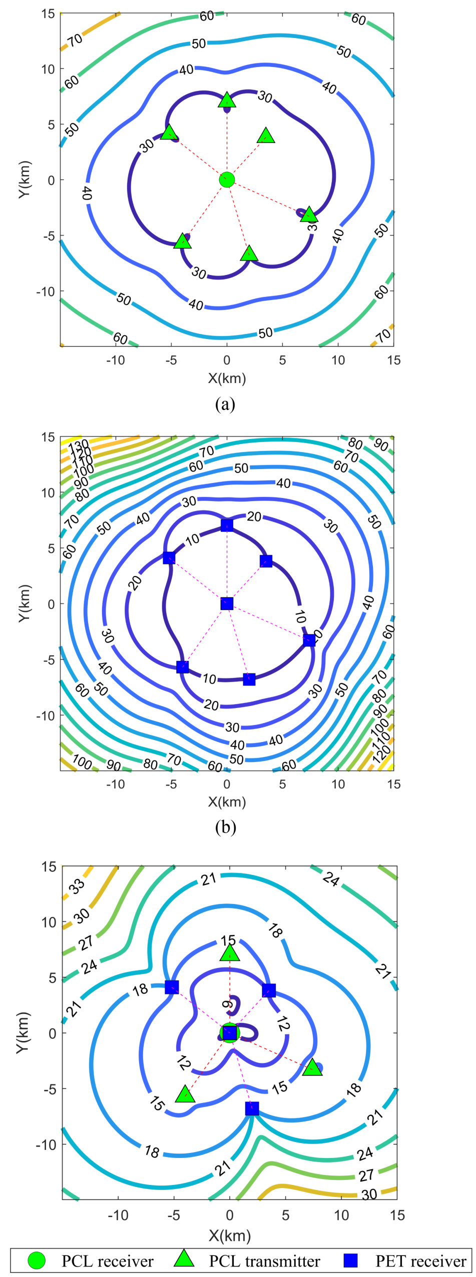
A. Comparison of Localization CRLB

Aubry et al. [17] plotted the contour lines of CRLB for the localization accuracy of the PCL-PET hybrid heterogeneous network and also theoretically analyzed that the Fisher information matrix of the hybrid heterogeneous network is larger than that of the homogeneous network. However, in the comparative analysis of [17], the number of measurements in the hybrid heterogeneous network is greater than that of the homogeneous network. In this section, we provide another perspective to argue the superiority of localization accuracy of the PCL-PET hybrid heterogeneous network.

Let the total number of measurements in the radar network be a constant value of 6, i.e., $N_{\mathrm{PCL}} + N_{\mathrm{PET}} = 6$. The standard deviation of PCL measurements is set as $\sigma_{\mathrm{PCL}} = 40$ m, and the standard deviation of PET measurements is set as $\sigma_{\mathrm{PET}} = 10$ m. Fig. 1 shows contour lines of localization accuracy under three different network configurations.

When $N_{\mathrm{PCL}} = 6$ and $N_{\mathrm{PET}} = 0$, it corresponds to a PCL network, with the positions of six transmitter–receiver pairs, as shown in Fig. 1(a). For the convenience of subsequent comparison, all receivers are set at the coordinate origin. It can be observed that the CRLB at (15, −15) km is approximately 70 m.

Setting $N_{\mathrm{PCL}} = 0$ and $N_{\mathrm{PET}} = 6$ yields a PET network, with PET node coordinates consistent with those in Fig. 1(a) and the reference node located at the coordinate origin. Fig. 1(b) shows the corresponding CRLB. It can be observed that the PET network exhibits high localization accuracy for nearby targets. However, as the target distance increases, the localization accuracy decreases rapidly, with the CRLB exceeding 120 m at (15, −15) km.



(c)

Fig. 1. CRLB for radar networks. (a) PCL network localization accuracy CRLB ($N_{\mathrm{PCL}} = 6$, $N_{\mathrm{PET}} = 0$, $\sigma_{\mathrm{PET}} = 40$ m). (b) PET network localization accuracy CRLB $N_{\mathrm{PCL}} = 0$, $N_{\mathrm{PET}} = 6$, $\sigma_{\mathrm{PET}} = 10$ m). (c) PCL-PET hybrid heterogeneous network localization accuracy CRLB ($N_{\mathrm{PCL}} = 3$, $N_{\mathrm{PET}} = 3$, $\sigma_{\mathrm{PCL}} = 40$ m, $\sigma_{\mathrm{PET}} = 10$ m).

Taking $N_{\mathrm{PCL}} = 3$ and $N_{\mathrm{PET}} = 3$ yields a PCL-PET hybrid heterogeneous network, where PCL receivers and PET reference nodes are at the coordinate origin. Fig. 1(c) shows the corresponding CRLB. For nearby targets, the localization accuracy of the hybrid heterogeneous network is almost comparable to the PET network and significantly better than the PCL network. For targets at (15, −15) km, the hybrid heterogeneous network shows a considerable

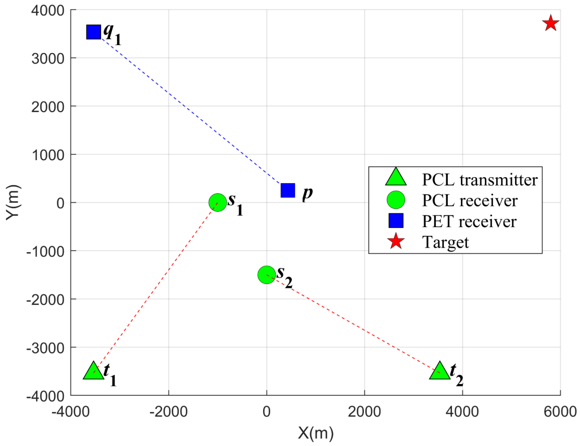


Fig. 2. Target localization scenario 1.

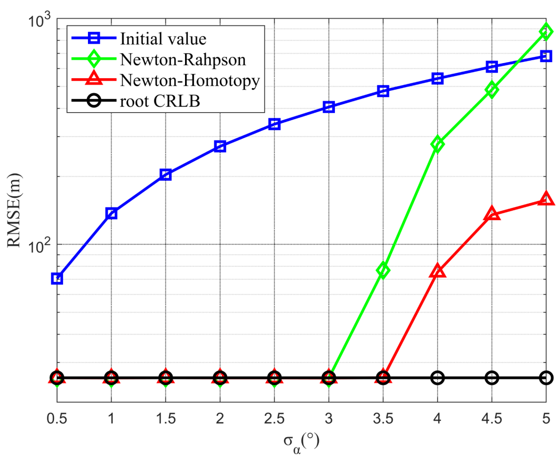


Fig. 3. Comparison of RMSE between Newton–Homotopy and Newton–Raphson.

improvement in localization accuracy compared to both PCL and PET networks. In conclusion, the PCL-PET hybrid heterogeneous network has advantages over homogeneous networks.

B. Target Localization (Scenario 1)

Consider the target localization scenario depicted in Fig. 2, which comprises two PCL receivers $\boldsymbol{s}_1$ and $\boldsymbol{s}_2$, two transmitters $\boldsymbol{t}_1$ and $\boldsymbol{t}_2$, and two PET nodes $\boldsymbol{q}_1$ and $\boldsymbol{p}$. The true value of the target position is denoted as $\boldsymbol{u}^0 = [5800, 3713]^{\mathrm{T}}$m. Only the measurements of pairs $\boldsymbol{s}_1 - \boldsymbol{t}_1$, $\boldsymbol{s}_2 - \boldsymbol{t}_2$, and $\boldsymbol{p} - \boldsymbol{q}_1$ contribute to the localization, resulting in $N_{\mathrm{PCL}} = 2$ and $N_{\mathrm{PET}} = 1$.

To compare the localization performance of the Newton–Homotopy and Newton–Raphson algorithms, we assume that the PCL receiver $\boldsymbol{s}_1$ can obtain not only the target's range measurement $\tilde{r}_{\mathrm{PCL},1}$ but also its azimuth angle measurement $\tilde{\alpha}_1$. Thus, the initial localization value can be determined by (43). Let us set $\sigma_{\mathrm{PCL}} = 40$ m and $\sigma_{\mathrm{PET}} = 10$ m. The value of n in s_n is set as 10, which means $\Delta s = s_i - s_{i-1} = 0.1 (i = 1, 2, \ldots, 10)$.

Fig. 3 illustrates the comparison of localization accuracy between the initial localization value and the two algorithms

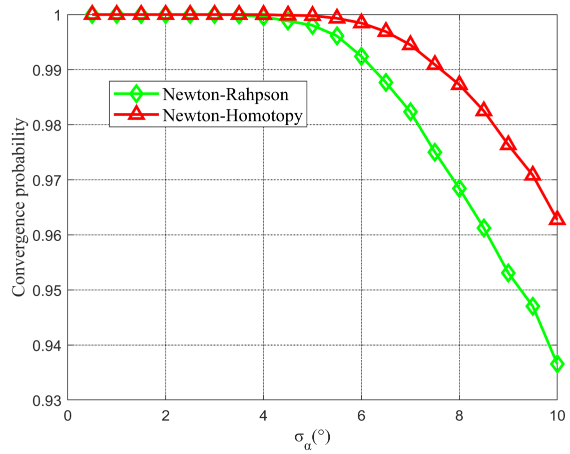


Fig. 4. Comparison of convergence probabilities between Newton–Homotopy and Newton–Raphson.

as the standard deviation σ_α of the azimuth angle $\tilde{\alpha}_1$ varies. The number of Monte Carlo simulations is 50 000.

Since only the value of σ_α is varied during the simulations, and the azimuth angle $\tilde{\alpha}_1$ is only used to calculate the initial localization result, the CRLB remains unchanged. It can be observed that when $\sigma_\alpha \leq 3^\circ$, both the Newton–Raphson algorithm and the Newton–Homotopy algorithm can achieve the CRLB. However, when $\sigma_\alpha = 3.5^\circ$, the RMSE of the initial value is already close to 500 m. The Newton–Raphson algorithm fails to reach the CRLB, while the Newton–Homotopy algorithm still achieves it. As σ_α continues to increase, the RMSE of the Newton–Raphson algorithm grows rapidly. When $\sigma_\alpha = 5^\circ$, the RMSE of the Newton–Raphson algorithm even surpasses the initial value. In contrast, the Newton–Homotopy algorithm does not exhibit such rapid degradation. When $\sigma_\alpha = 5^\circ$, the RMSE of the Newton–Homotopy is only 150 m.

In Fig. 4, the convergence probabilities of the two algorithms are compared. When the determinant of the Hessian matrix is less than 4×10^{-5} during the iteration process, the algorithm will be considered to have diverged. From Fig. 3, it can be observed that when $\sigma_\alpha \leq 3^\circ$, the convergence probability of both algorithms is 1. As σ_α increases, the Newton–Raphson algorithm exhibits earlier divergence compared to the Newton–Homotopy algorithm, and the convergence probability of the Newton–Raphson algorithm decreases more rapidly. The convergence probability curve in Fig. 4 corroborates with the RMSE comparison in Fig. 3, indicating that the Newton–Homotopy algorithm outperforms the Newton–Raphson algorithm in large initial error.

Setting $\sigma_\alpha = 3.5^\circ$, $\sigma_{\mathrm{PCL}} = \sigma$, $\sigma_{\mathrm{PET}} = 0.25\sigma$, the two localization algorithms proposed in this article are verified by varying the value of σ. The simulation scenario setup remains as depicted in Fig. 2.

Fig. 5 illustrates the comparison of localization accuracy from 5000 Monte Carlo simulations. The RMSE of the SDP increases with the increment of σ, displaying an approximately linear correspondence. The "SDP+Improvement" is the result of (42), which performs error correction on the SDP. It can be seen that the effect of error correction is

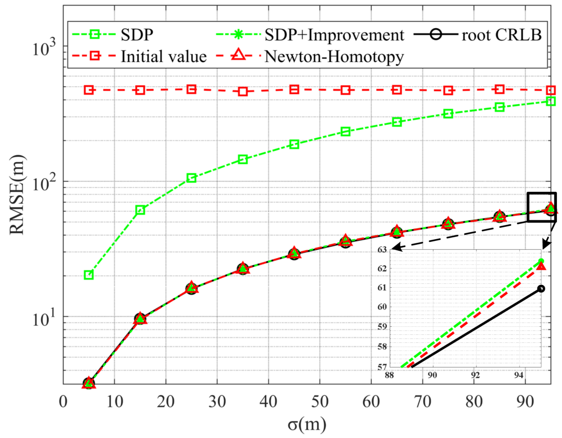


Fig. 5. Target localization RMSE of the proposed algorithms.

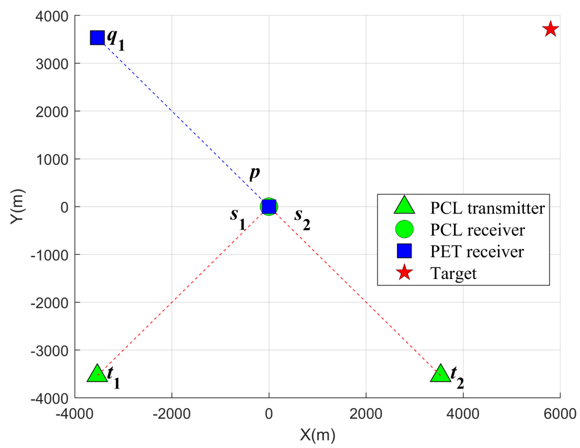


Fig. 6. Target localization scenario 2.

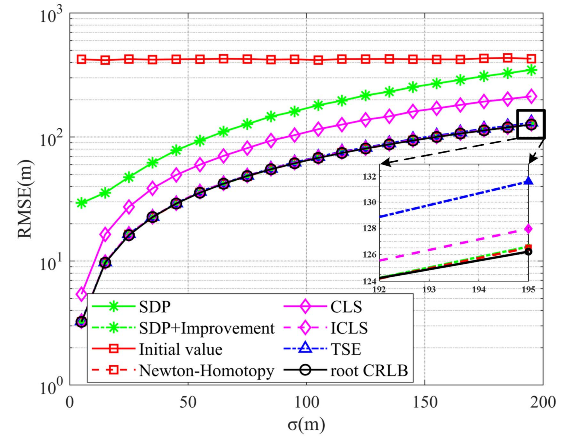


Fig. 7. Comparsion of RMSEs using different algorithms in the scenario 2.

significant, and the localization accuracy can reach CRLB when $\sigma \leq 85$ m. The Initial value is obtained through (43). Due to the dominance of azimuth angle error σ_α, its RMSE hardly changes with the increase in σ. The RMSE of the Newton–Homotopy can also reach the CRLB. From the subplot in Fig. 5, it can be seen that when $\sigma = 95$ m, the two proposed algorithms still exhibit high localization accuracy, with a difference from the CRLB of only approximately 1 m.

The average computational time for the "SDP+Improvement" is 0.2943 s, while for the Newton–Homotopy, it is only 0.0026 s. Therefore, when the initial localization result is obtainable, it is recommended to utilize the Newton–Homotopy algorithm for faster computational speed.

C. Target Localization (Scenario 2)

In this section, the proposed algorithms will be compared with existing algorithms, including TSE, CLS, and ICLS. Considering that existing algorithms are only applicable to situations where the PCL and PET receivers have the same location, $\boldsymbol{s}_1 = \boldsymbol{s}_2 = \boldsymbol{p} = [0, 0]^{\mathrm{T}}$ is set in scenario 2, while other simulation parameters remain consistent with scenario 1. Figs. 6 and 7 respectively show the simulation scenario and results.

TABLE I
Positions of the Three Targets

Target	#1	#2	#3
Position(m)	(5800, 3713)	(0, 5000)	(2500,1500)

It can be seen that when the measurement error level is low, TSE, ICLS, and the proposed algorithms can all achieve CRLB. Due to ignoring the covariance matrix of measurement errors and not performing iterative operations, the RMSE of the CLS did not reach CRLB. When the measurement error increases to $\sigma = 195$ m, both the TSE and ICLS have deviated from CRLB, with deviations of approximately 5 and 2 m. In comparison, the deviations between the two proposed algorithms and CRLB is the smallest, only 0.4 m. The comparsion result indicates that the proposed algorithms have excellent localization accuracy. As there is no need to limit the location of the receivers, it means that the proposed algorithms have strong generality.

D. Measurement Association

In Section III-C, we derived that the test statistic approximately follows a chi-square distribution. This section conducts simulation verification on the correctness of the derivation.

Considering that measurement errors and target position mainly influence the approximation effect, three targets with different positions are set in this simulation. The positions of the three targets are provided in Table I. The PCL-PET hybrid heterogeneous network configuration remains consistent with the previous section. Therefore, we need to verify whether the test statistic follows a chi-square distribution with a degree of freedom of 1 ($N_{\mathrm{PCL}} + N_{\mathrm{PET}} - N_{\mathrm{D}} = 1$).

Setting $\sigma_\alpha = 3.5^\circ$, $\sigma_{\mathrm{PCL}} = \sigma$, $\sigma_{\mathrm{PET}} = 0.25\sigma$, and the Newton–Homotopy algorithm is used for localization. We use the Kolmogorov–Smirnov (K–S) test method to do the verification. Specifically, by comparing the K–S statistic with the critical value at a given significance level, if the K–S statistic is less than the critical value, we accept that the sample follows the distribution; otherwise, we reject it.

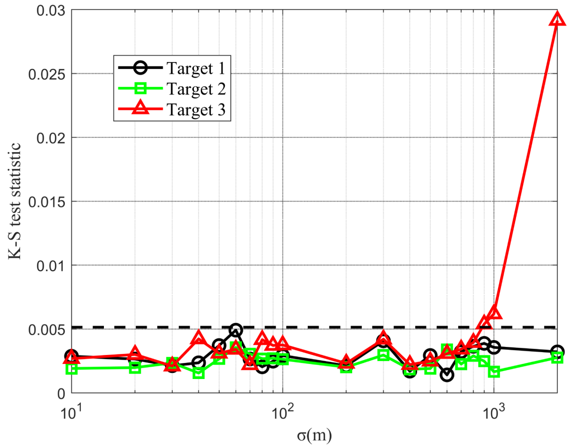


Fig. 8. Verification results of proposed test statistics (the black horizontal dashed line represents the critical value at a significance level of 0.01).

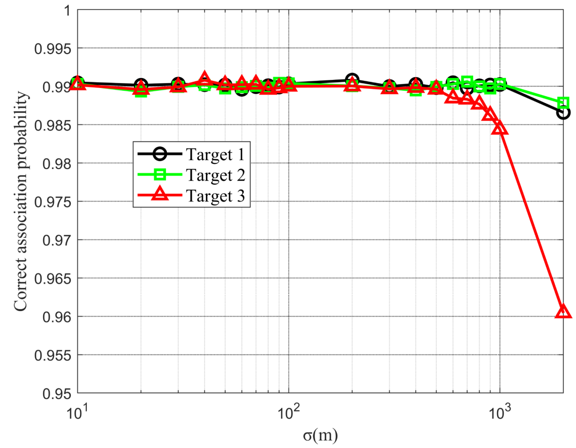


Fig. 9. Correct association probability of three targets.

For a given measurement error level and each target, 100 000 Monte Carlo simulations are conducted.

Fig. 8 displays the K–S statistics corresponding to three targets. A total of 20 levels of measurement error are simulated. The significance level is set as 0.01, with the corresponding critical value being approximately 0.005, marked by a black horizontal dashed line in Fig. 8. It can be observed that when $\sigma \leq 10^3$m, the K–S statistics for all three targets are below the critical value, leading to the conclusion that the test statistic follows a chi-square distribution with 1 degree of freedom. However, when $\sigma = 2 \times 10^3$m, we notice that the K–S statistic for Target 3 exceeds the critical value, indicating a deviation of the corresponding test statistic from the chi-square distribution. This is attributed to the fact that, compared to Target 1 and 2, the localization accuracy of Target 3 deteriorates more rapidly with increasing σ, resulting in larger approximation errors.

We further validate the correct association probability in practical applications through simulation. Taking $P_{\mathrm{A}} = 0.99$, the corresponding decision threshold $\lambda \approx 6.635$. When the test statistic is less than 6.635, we accept the corresponding measurement association hypothesis; otherwise, we reject it. Let the number of times the correct association is accepted be represented by N_{A}, and the number of Monte Carlo simulations be represented by N_{MC} ($N_{\mathrm{MC}} = 100\,000$). Therefore, the actual correct association probability can be calculated as $\bar{P}_{\mathrm{A}} = N_{\mathrm{A}}/N_{\mathrm{MC}}$. Fig. 9 illustrates the correct association probability for the three targets.

It can be observed that when $\sigma < 1000$ m, the correct association probabilities for all three targets remain consistent with the preset value $P_{\mathrm{A}} = 0.99$, indicating that the proposed algorithm not only effectively judges the measurement interconnection hypothesis but also provides a reference for selecting the decision threshold. However, when $\sigma \geq 1000$ m, the correct association probability for the Target 3 deviates from the preset value. This is consistent with Fig. 6 and is attributed to the larger localization error. Both Figs. 8 and 9 indicate that the proposed measurement association algorithm is applicable to a large range of measurement errors.

V. CONCLUSION

In this article, we discuss the problem of target localization and measurement association in a PCL-PET hybrid heterogeneous network. First, by plotting the CRLB contour lines for localization accuracy, we have demonstrated that the hybrid heterogeneous network can achieve higher localization accuracy than single-sensor networks. Subsequently, addressing the general scenario where PCL receivers can only receive signals from specific transmitters, we propose two localization algorithms. The SDP-based algorithm can run in situations where the initial value of the target position is not available at the cost of high computational complexity; the Newton–Homotopy-based algorithm requires an initial value as a driver, which has a higher convergence probability and lower computational complexity. The simulation verified that the accuracy of both algorithms can reach CRLB. Regarding the measurement association problem, we propose an algorithm for constructing the test statistic. Simulation results indicate that the proposed test statistic follows the chi-square distribution, and the correct association probability is almost consistent with the preset value. Future work will discuss closed-form solutions to the target localization problem in a PCL-PET hybrid heterogeneous network, aiming to achieve higher computational efficiency.

APPENDIX

This section considers a two-dimensional target localization scenario and provides detailed expressions for $\boldsymbol{H}(\boldsymbol{u}, s_i)$ and $\boldsymbol{h}(\boldsymbol{u}, s_i)$ in the Newton–Homotopy algorithm. It is assumed that $\boldsymbol{u} = [x, y]^{\mathrm{T}}$, $\boldsymbol{s}_i = [x_i^{(s)}, y_i^{(s)}]^{\mathrm{T}}$, $\boldsymbol{t}_i = [x_i^{(t)}, y_i^{(t)}]^{\mathrm{T}}$, $\boldsymbol{q}_j = [x_j^{(q)}, y_j^{(q)}]^{\mathrm{T}}$, $\boldsymbol{p} = [x^{(p)}, y^{(p)}]^{\mathrm{T}}$, $i = 1, 2, \ldots, N_{\mathrm{PCL}}$, and $j = 1, 2, \ldots, N_{\mathrm{PET}}$.

Substituting (48) into (51) yields

$$\boldsymbol{h}(\boldsymbol{u}, s_i) = \nabla f(\boldsymbol{u}) + (s_i - 1) \nabla f(\hat{\boldsymbol{u}})$$

$$
\begin{aligned}
&= \frac{\partial f(\boldsymbol{u})}{\partial \boldsymbol{u}} + (s_i - 1)\left.\frac{\partial f(\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\hat{\boldsymbol{u}}} \\
&= 2\frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial \boldsymbol{u}}\boldsymbol{Q}^{-1}\left[\boldsymbol{g}(\boldsymbol{u}) - \tilde{\boldsymbol{r}}\right] \\
&\quad + 2(s_i - 1)\left.\frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\hat{\boldsymbol{u}}}\boldsymbol{Q}^{-1}\left[\boldsymbol{g}(\hat{\boldsymbol{u}}) - \tilde{\boldsymbol{r}}\right] \qquad (69)
\end{aligned}
$$

where $\frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial \boldsymbol{u}}$ is the first-order partial derivative of the measurement vector with respect to the target position

$$
\frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial \boldsymbol{u}} = \begin{bmatrix}
\frac{(\boldsymbol{u}-\boldsymbol{t}_1)^{\mathrm{T}}}{\|\boldsymbol{u}-\boldsymbol{t}_1\|} + \frac{(\boldsymbol{u}-\boldsymbol{s}_1)^{\mathrm{T}}}{\|\boldsymbol{u}-\boldsymbol{s}_1\|} \\
\vdots \\
\frac{(\boldsymbol{u}-\boldsymbol{t}_{N_{\mathrm{PCL}}})^{\mathrm{T}}}{\|\boldsymbol{u}-\boldsymbol{t}_{N_{\mathrm{PCL}}}\|} + \frac{(\boldsymbol{u}-\boldsymbol{s}_{N_{\mathrm{PCL}}})^{\mathrm{T}}}{\|\boldsymbol{u}-\boldsymbol{s}_{N_{\mathrm{PCL}}}\|} \\
\frac{(\boldsymbol{u}-\boldsymbol{q}_1)^{\mathrm{T}}}{\|\boldsymbol{u}-\boldsymbol{q}_1\|} - \frac{(\boldsymbol{u}-\boldsymbol{p})^{\mathrm{T}}}{\|\boldsymbol{u}-\boldsymbol{p}\|} \\
\vdots \\
\frac{(\boldsymbol{u}-\boldsymbol{q}_{N_{\mathrm{PET}}})^{\mathrm{T}}}{\|\boldsymbol{u}-\boldsymbol{q}_{N_{\mathrm{PET}}}\|} - \frac{(\boldsymbol{u}-\boldsymbol{p})^{\mathrm{T}}}{\|\boldsymbol{u}-\boldsymbol{p}\|}
\end{bmatrix}^{\mathrm{T}}. \qquad (70)
$$

Substituting (69) into (53) yields

$$
\begin{aligned}
\boldsymbol{H}(\boldsymbol{u}, s_i) &= \frac{\partial \boldsymbol{h}(\boldsymbol{u}, s_i)}{\partial \boldsymbol{u}^{\mathrm{T}}} = \frac{\partial \nabla f(\boldsymbol{u})}{\partial \boldsymbol{u}^{\mathrm{T}}} + 0 \\
&= \frac{\partial^2 f(\boldsymbol{u})}{\partial \boldsymbol{u}\partial \boldsymbol{u}^{\mathrm{T}}} = \begin{bmatrix} \frac{\partial^2 f(\boldsymbol{u})}{\partial x \partial x} & \frac{\partial^2 f(\boldsymbol{u})}{\partial x \partial y} \\ \frac{\partial^2 f(\boldsymbol{u})}{\partial y \partial x} & \frac{\partial^2 f(\boldsymbol{u})}{\partial y \partial y} \end{bmatrix}. \qquad (71)
\end{aligned}
$$

where

$$
\begin{aligned}
\frac{\partial^2 f(\boldsymbol{u})}{\partial x \partial x} &= 2\frac{\partial^2 \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial x \partial x}\boldsymbol{Q}^{-1}\left[\boldsymbol{g}(\boldsymbol{u}) - \tilde{\boldsymbol{r}}\right] \\
&\quad + 2\frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial x}\boldsymbol{Q}^{-1}\frac{\partial \boldsymbol{g}(\boldsymbol{u})}{\partial x} \qquad (72) \\
\frac{\partial^2 f(\boldsymbol{u})}{\partial y \partial y} &= 2\frac{\partial^2 \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial y \partial y}\boldsymbol{Q}^{-1}\left[\boldsymbol{g}(\boldsymbol{u}) - \tilde{\boldsymbol{r}}\right] \\
&\quad + 2\frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial y}\boldsymbol{Q}^{-1}\frac{\partial \boldsymbol{g}(\boldsymbol{u})}{\partial y} \qquad (73) \\
\frac{\partial^2 f(\boldsymbol{u})}{\partial x \partial y} &= \frac{\partial^2 f(\boldsymbol{u})}{\partial y \partial x} = 2\frac{\partial^2 \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial x \partial y}\boldsymbol{Q}^{-1}\left[\boldsymbol{g}(\boldsymbol{u}) - \tilde{\boldsymbol{r}}\right] \\
&\quad + 2\frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial x}\boldsymbol{Q}^{-1}\frac{\partial \boldsymbol{g}(\boldsymbol{u})}{\partial y}. \qquad (74)
\end{aligned}
$$

$\frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial x}$ and $\frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial y}$ are the first-order partial derivatives of $\boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})$, correspond to the first and second rows of the matrix $\frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial \boldsymbol{u}}$, respectively. The expressions for the second-order derivatives of $\boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})$ are as follows:

$$
\frac{\partial^2 \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial x \partial x} = \left[\boldsymbol{g}_1^{\mathrm{T}}, \boldsymbol{g}_2^{\mathrm{T}}\right] \qquad (75)
$$

$$
\frac{\partial^2 \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial y \partial y} = \left[\boldsymbol{g}_3^{\mathrm{T}}, \boldsymbol{g}_4^{\mathrm{T}}\right] \qquad (76)
$$

$$
\frac{\partial^2 \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{u})}{\partial x \partial y} = \left[\boldsymbol{g}_5^{\mathrm{T}}, \boldsymbol{g}_6^{\mathrm{T}}\right] \qquad (77)
$$

$$
\left[\boldsymbol{g}_1\right]_i = \frac{\|\boldsymbol{u}-\boldsymbol{t}_i\|^2 - \left(x - x_i^{(t)}\right)^2}{\|\boldsymbol{u}-\boldsymbol{t}_i\|^3} + \frac{\|\boldsymbol{u}-\boldsymbol{s}_i\|^2 - \left(x - x_i^{(s)}\right)^2}{\|\boldsymbol{u}-\boldsymbol{s}_i\|^3} \qquad (78)
$$

$$
\left[\boldsymbol{g}_2\right]_j = \frac{\|\boldsymbol{u}-\boldsymbol{q}_j\|^2 - \left(x - x_j^{(q)}\right)^2}{\|\boldsymbol{u}-\boldsymbol{q}_j\|^3} - \frac{\|\boldsymbol{u}-\boldsymbol{p}\|^2 - \left(x - x^{(p)}\right)^2}{\|\boldsymbol{u}-\boldsymbol{p}\|^3} \qquad (79)
$$

$$
\left[\boldsymbol{g}_3\right]_i = \frac{\|\boldsymbol{u}-\boldsymbol{t}_i\|^2 - \left(y - y_i^{(t)}\right)^2}{\|\boldsymbol{u}-\boldsymbol{t}_i\|^3} + \frac{\|\boldsymbol{u}-\boldsymbol{s}_i\|^2 - \left(y - y_i^{(s)}\right)^2}{\|\boldsymbol{u}-\boldsymbol{s}_i\|^3} \qquad (80)
$$

$$
\left[\boldsymbol{g}_4\right]_j = \frac{\|\boldsymbol{u}-\boldsymbol{q}_j\|^2 - \left(y - y_j^{(q)}\right)^2}{\|\boldsymbol{u}-\boldsymbol{q}_j\|^3} - \frac{\|\boldsymbol{u}-\boldsymbol{p}\|^2 - \left(y - y^{(p)}\right)^2}{\|\boldsymbol{u}-\boldsymbol{p}\|^3} \qquad (81)
$$

$$
\left[\boldsymbol{g}_5\right]_i = -\frac{\left(x - x_i^{(t)}\right)\left(y - y_i^{(t)}\right)}{\|\boldsymbol{u}-\boldsymbol{t}_i\|^3} - \frac{\left(x - x_i^{(s)}\right)\left(y - y_i^{(s)}\right)}{\|\boldsymbol{u}-\boldsymbol{s}_i\|^3} \qquad (82)
$$

$$
\left[\boldsymbol{g}_6\right]_j = -\frac{\left(x - x_j^{(q)}\right)\left(y - y_j^{(q)}\right)}{\|\boldsymbol{u}-\boldsymbol{q}_j\|^3} + \frac{\left(x - x^{(p)}\right)\left(y - y^{(p)}\right)}{\|\boldsymbol{u}-\boldsymbol{p}\|^3}. \qquad (83)
$$

At this point, the expressions for $\boldsymbol{H}(\boldsymbol{u}, s_i)$ and $\boldsymbol{h}(\boldsymbol{u}, s_i)$ have been provided in full. In practical applications, substituting the value of $\hat{\boldsymbol{u}}(s_i)$ into (69) and (71) is all that is needed.

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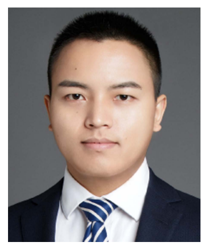

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